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PowerFactory 2024

Technical Reference

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1 General Description

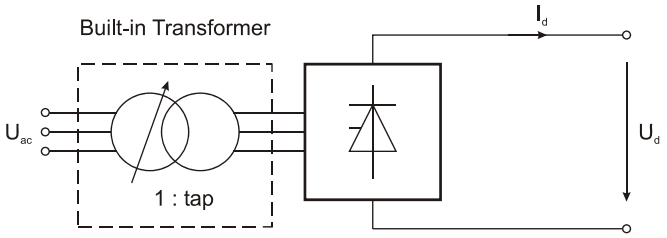


Figure 1.1: HVDC LCC converter including built-in transformer

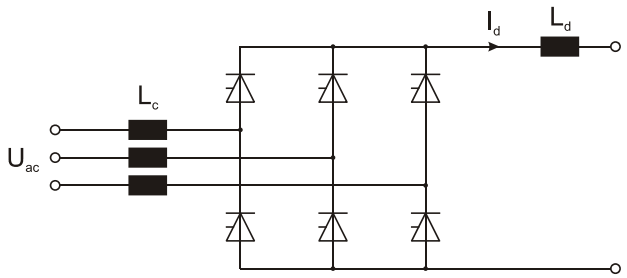


Figure 1.2: Detailed LCC circuit with commutation reactance and DC reactance (not part of the model)

The model can be configured as:

- Three-phase diode rectifier
- Three-phase line-commutated rectifier/inverter
- Three-phase capacitor-commutated rectifier/inverter

The orientation of the element has to be specified to allow representation of either a rectifier or an inverter.

The diode rectifier is a full-bridge diode rectifier, which is rectifying the three-phase AC voltage to a 6-pulse DC voltage. Due to the usage of diodes, which can neither be turned-on or turned-off externally, the DC voltage or DC current of the rectifier cannot be controlled.

The controlled line-commutated converter (LCC) consists of six power thyristors, arranged as shown in Figure 1.2. These valves can be turned-on by an external control signal, but only turn-off when the current flowing through them becomes negative. This converter can operate as rectifier or as inverter, depending on the timing of the gate signal relative to the AC voltage wave.

The controlled capacitor-commutated converter (CCC) uses capacitors between the thyristors and the converter transformer. The capacitors result in an advance of the commutating voltage over the thyristors, allowing to operate the converter with lower firing angle thus drawing lower reactive power from the grid. A capacitor-commutated inverter is also less susceptible to commutation failure during grid faults. The CCC configuration is shown in Figure 1.3 and the detailed circuit in Figure 1.4. The built-in transformer is optional.

Fundamental frequency model are used for load flow calculations and stability analysis, and are described in sections 1.1.1 and 1.1.2. The detailed modelling of all six thyristors is only

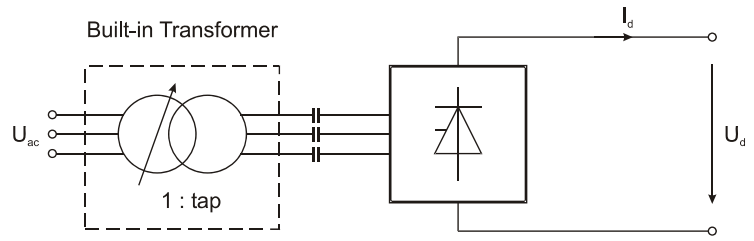


Figure 1.3: HVDC CCC converter including built-in transformer)

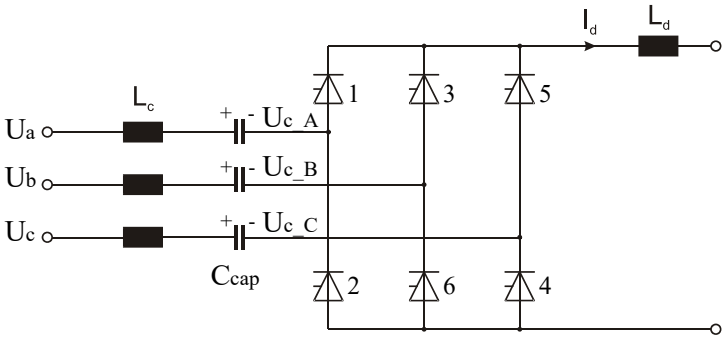


Figure 1.4: Detailed CCC circuit with commutation reactance and DC reactance (not part of the model)

necessary for EMT simulations, where the converters are modelled as shown in Figures 1.2 and 1.4.

1.1 Fundamental Frequency Model

The models for load flow calculation and RMS-simulation are based on a fundamental frequency approach.

1.1.1 Line-commutated converter

The equations of the thyristor converter and the diode rectifier are identical if the diode rectifier is assumed as an uncontrolled thyristor converter (hence the firing angle α is set to zero).

During steady-state operation the converter can be modelled as a load with constant active and reactive power *P* and *Q*. The following equations describe the converter in a detailed way and give hints for the layout of an HVDC system.

The transmitted DC power of the high-voltage DC system is given by:

$$P_d = U_d \cdot I_d \tag{1}$$

The DC voltage of the ideal and uncontrolled converter, without load, is called the “ideal no-load direct voltage” U_{d0} , which is defined as follows:

$$U_{d0} = \frac{s_0 \cdot q}{\pi} \cdot \sin\left(\frac{\pi}{q}\right) \cdot \frac{\sqrt{2}}{\sqrt{3}} \cdot U_{LL} \tag{2}$$

where s_0 defines the number of commutation groups, q is the number of branches in a commutation group and U_{LL} is the AC voltage supplied to the converter station. In case the converter is connected to an external transformer U_{LL} is calculated according to Equation 32. For a 6-pulse converter there are two commutation groups ($s_0 = 2$) and q is equal to 3, hence the U_{d0} is

$$U_{d0} = \frac{3 \cdot \sqrt{2}}{\pi} U_{LL} \approx 1.35 \cdot U_{LL} \tag{3}$$

This equation is valid for the uncontrolled thyristor converter ($\alpha = 0$) as well as for the diode bridge. The gate control of the thyristors can be used to delay the ignition of the valves. The time delay due to the turn-on signal applied is defined to be $\omega t = \alpha$. Then the DC voltage depends on the ignition angle α

$$U_{d\alpha} = U_{d0} \cdot \cos(\alpha) \tag{4}$$

The effect of the ignition angle α is shown in Figure 1.5, where the AC voltage, the phase currents and the DC voltage can be seen for an idealized operation with the DC current I_d assumed to be constant. The ignition angle is also indicated in the figure. Here the time between the transfer of the current from one valve to the next is assumed to be zero, i.e the leakage reactance of the transformer is neglected and the commutation angle μ is zero.

To study more realistic converters the current commutation from one valve to the next must be considered. The commutation leads to a drop in the DC voltage ΔU_d:

$$U_d = U_{d0} \cdot \cos(\alpha) - \Delta U_d \tag{5}$$

ΔU_d is defined as a function of I_d and the commutation reactance X_c according to equation 6, where R_{cr} is the "equivalent commutating resistance". R_{cr} does not represent a real resistance and thus has no associated power losses. ΔU_d is defined as:

$$\Delta U_d = -R_{cr} \cdot I_d = -\frac{3}{\pi} \cdot \omega \cdot L_c \cdot I_d = -\frac{3}{\pi} \cdot X_c \cdot I_d \tag{6}$$

Combining equations 5 and 6, the DC voltage can be expressed as:

$$U_d = U_{d0} \cdot \cos(\alpha) + \frac{3}{\pi} \cdot X_c \cdot I_d \tag{7}$$

Figure 1.6 shows the equivalent circuit for the rectifier including the effects of commutation. Note in the figure and in the equations above, that the DC current is negative for the rectifier operation due to the representation with load-orientation.

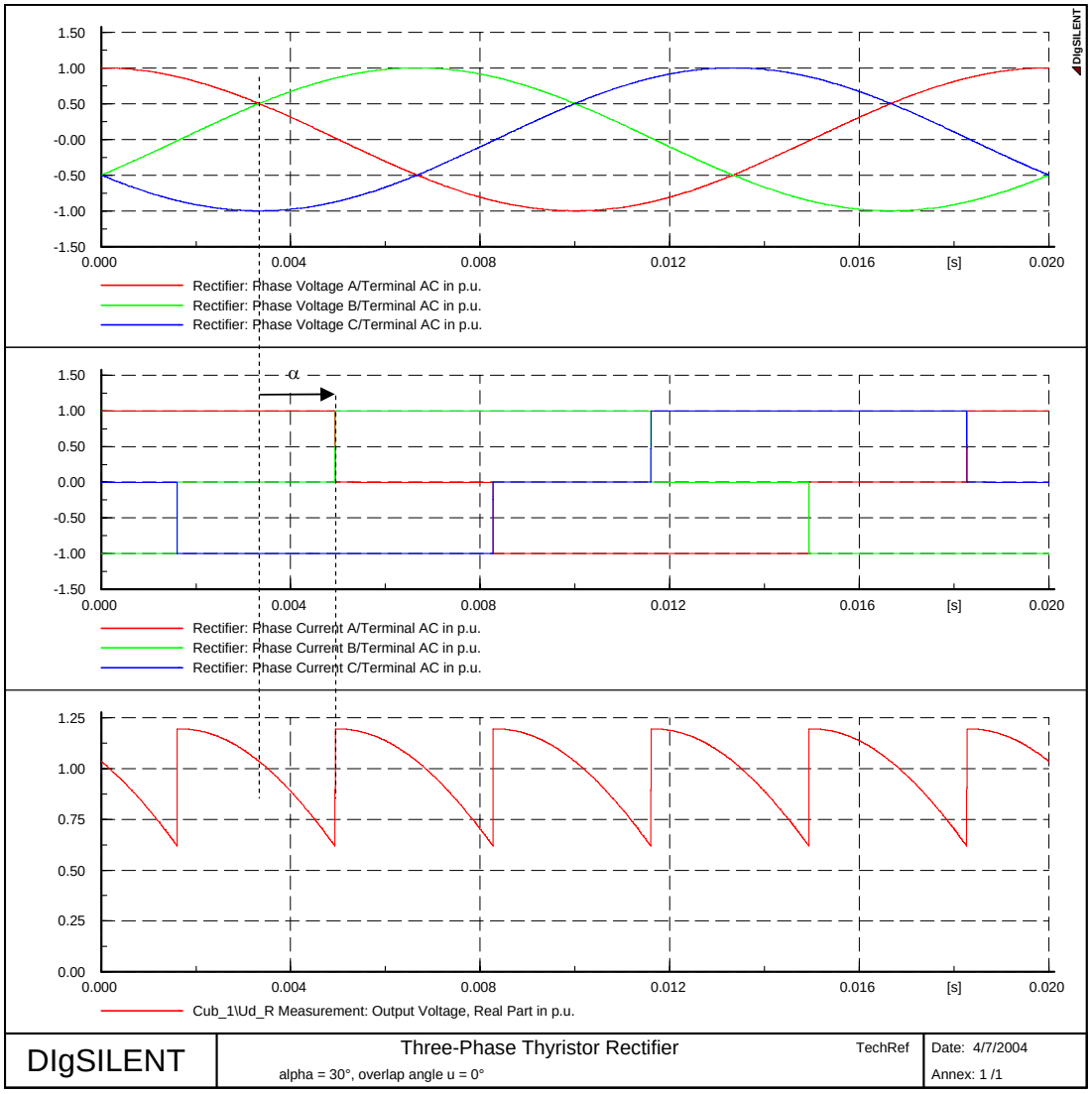


Figure 1.5: Phase voltages, phase currents and DC voltage of a three-phase rectifier operating with $\alpha = 30°$ and zero commutation angle μ

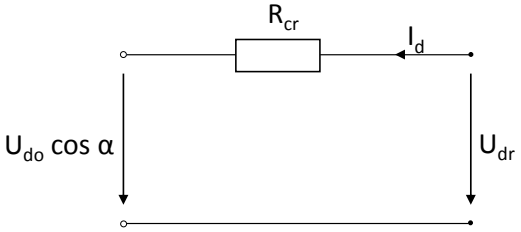


Figure 1.6: Rectifier equivalent circuit with equivalent commutating resistance R_{cr}

With the DC current being equal to its rated value I_d, the DC voltage can be written in a different form as:

$$U_d = U_{d0} \cdot (\cos(\alpha) - d_{xr}) \tag{8}$$

with

$$d_{xr} = \frac{1}{2} \cdot \frac{u_{kr}}{100} \tag{9}$$

The term d_{xr} has been calculated from the following relationship:

$$d_{xr} = \frac{3}{\pi} \cdot X_c \cdot \frac{|I_d|}{U_{d0}} \tag{10}$$

assuming the converter transformer rated current I_r is equal to $sqrt(2)/sqrt(3) \cdot I_d$ (see section 1.1.4) and expressing U_{do} according to equation 3.

The phase voltages and currents, as well as the DC voltage of a thyristor rectifier, including effects of commutation can be seen in Figure 1.7. The ignition angle α and the commutation angle μ are indicated in the figure.

Using these two angles two other angles can be defined. The extinction angle γ, which is normally used in the control on the inverter side of the HVDC, is defined as:

$$\gamma = \pi - \alpha - \mu \tag{11}$$

The ignition advance angle β is defined as

$$\beta = \pi - \alpha \tag{12}$$

β is often used in the HVDC controllers for both the rectifier and inverter side.

Using these different angles the DC voltage can be calculated differently for the rectifier and for the inverter respectively:

$$U_{dr} = U_{d0} \cdot \frac{\cos(\alpha) + \cos(\alpha + \mu)}{2} \tag{13}$$

and

$$U_{di} = U_{d0} \cdot \frac{\cos(\beta) + \cos(\gamma)}{2} \tag{14}$$

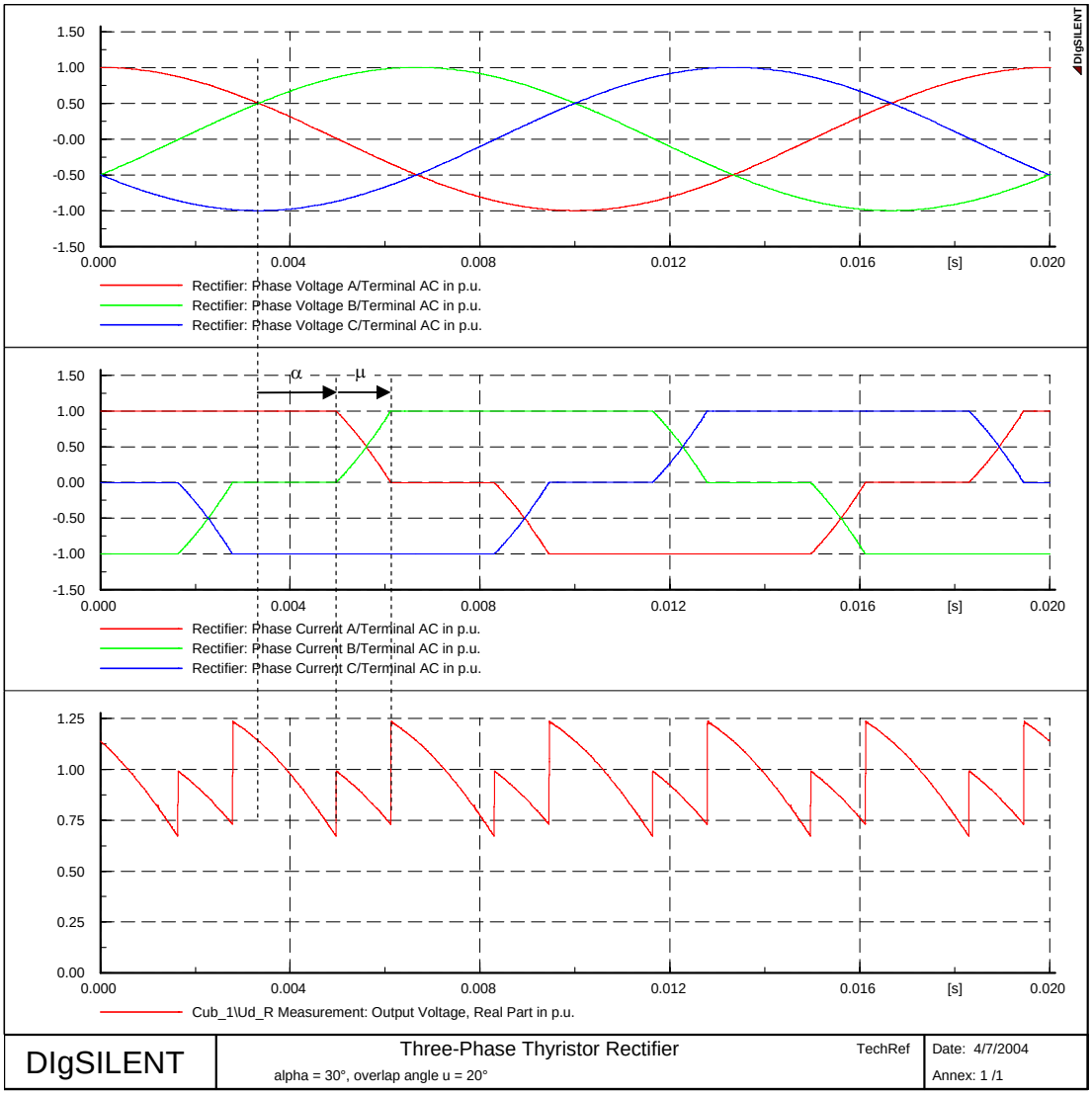


Figure 1.7: Phase voltages, phase currents and DC voltage of a three-phase rectifier operating with $\alpha = 30°$ and an overlap angle of $\mu = 20°$

The DC voltage for the inverter case is considered positive in equation 14. Using equations 13 and 8, the term d_{xr} can be expressed as:

$$d_{xr} = \frac{\cos(\alpha) - \cos(\alpha + \mu)}{2} \tag{15}$$

The phase currents of the 6-pulse bridge are shown in Figure 1.5 and Figure 1.7. In the literature the AC current is often calculated approximately from the ideal rectifier current with the commutation angle neglected. In *PowerFactory* the amplitude of the fundamental frequency current I_{L1} is calculated using the Fourier analysis of the phase current waveform, so the effect of the commutation is taken into account. This leads to the following relationship between the RMS value of the fundamental frequency component and the direct current:

$$I_{L1} = k \cdot \frac{\sqrt{6}}{\pi} \cdot I_d \tag{16}$$

where k is equal to

$$k = \frac{\sqrt{[\cos(2\alpha) - \cos 2(\alpha + \mu)]^2 + [2\mu + \sin(2\alpha) - \cos 2(\alpha + \mu)]^2}}{4 \cdot [\cos(\alpha) - \cos(\alpha + \mu)]} \tag{17}$$

This factor is close to unity for small values of μ, but if the angle becomes larger, the error increases up to 4% at $\mu = 60°$. For unsymmetrical operation the phase currents have to be calculated differently, which is described in section 1.1.6.

The power factor $cos(\varphi)$ can then be calculated, given the power equivalence on the AC and DC side and using equations 3, 14 and 17:

$$\sqrt{3} \cdot U_{LL} \cdot I_{L1} \cdot \cos(\varphi) = U_d \cdot \mathrm{I}_d = \frac{3\sqrt{2}}{\pi} \cdot U_{LL} \cdot \frac{\cos(\alpha) + \cos(\alpha + \mu)}{2} \cdot I_d \tag{18}$$

$$\cos(\varphi) = \frac{1}{2k} \cdot [\cos(\alpha) + \cos(\alpha + \mu)] \tag{19}$$

1.1.2 Capacitor-commutated converter

The capacitor commutated converter uses series capacitors between the converter transformer and the valves. It can be defined by selecting the option in the Basic Data page of *TypRec* type.

The operating voltage and current of a CCC rectifier are shown in Figure 1.8. At the moment when commutation between valve 1 and 3 is started, it can be seen that the effect of the capacitors is to advance the voltage $U_{v3} - U_{v1}$ over the valves in comparison to the line-line commutating voltage $U_b - U_a$. The firing angle α is the angle corresponding to the time between the $U_b - U_a$ voltage zero crossing and the gate triggering of valve 3. Since the voltage over the valves $U_{v3} - U_{v1}$ is positive before the line-line commutating voltage $U_b - U_a$ becomes positive, the firing angle α can be reduced to zero or even be negative without compromising the commutation process. As a result the reactive power requirement of the CCC is lower than an LCC. For negative firing angles, the CCC produces reactive power. Notice that the valve voltage U_{v1} is ideally zero when valve 1 is conducting.

The firing margin α' for a CCC rectifier is defined as the angle corresponding to the time between valve voltage zero crossing and valve gate triggering.

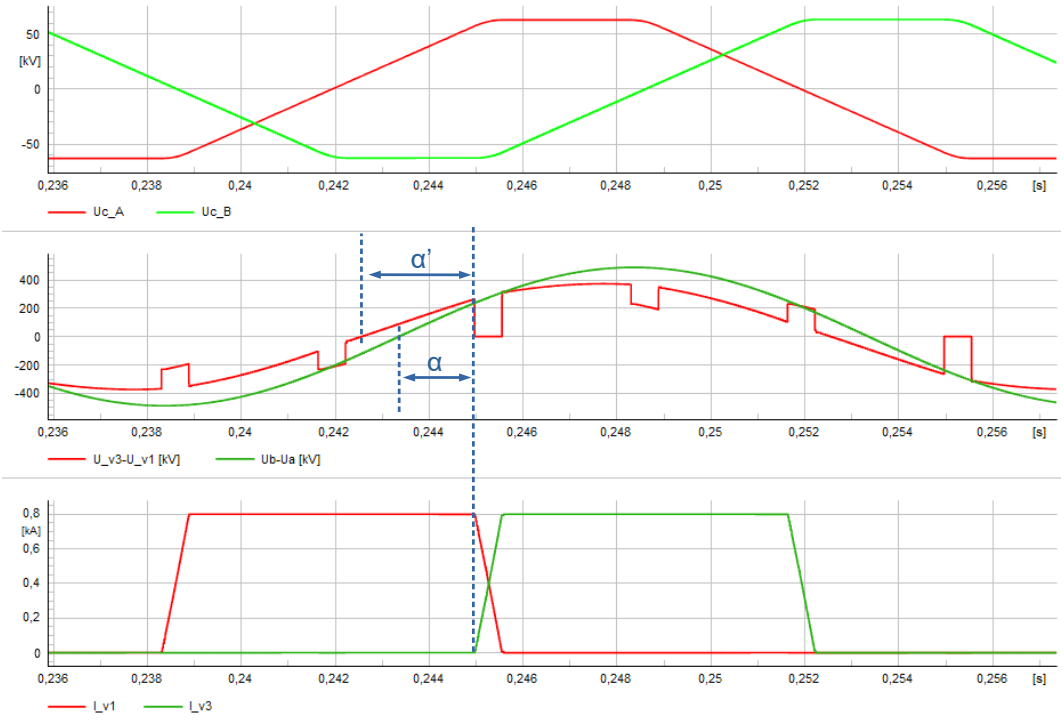


Figure 1.8: Capacitor voltages Uc_A and Uc_B, commutating voltage $U_b - U_a$, valve voltage $U_{v3} - U_{v1}$ and valve currents I_{v1} and I_{v3} for a CCC rectifier.

Analogously, for a CCC inverter the extinction angle γ can be reduced (or even be negative) compared to an LCC. With reference to the commutation process between valve 1 and valve 3, the commutation margin γ' is defined as the angle corresponding to the time between the instant when the outgoing valve current is extinguished and the valves voltage $U_{v1} - U_{v3}$ negative zero crossing, see Figure 1.9. In order to avoid commutation failure, the commutation margin must be positive. A CCC inverter typically controls the commutation margin.

The equations for the CCC are developed under the assumption of balanced conditions in the grid and are based on the models presented in [4] and [3].

$$U_{di0} = \frac{3\sqrt{2}}{\pi} \cdot U_{c,LL} \qquad d_{xl} = \frac{3}{\pi} \cdot X_c \qquad d_{xc} = \frac{3}{\pi \cdot \omega \cdot C} \qquad L_c = \frac{X_c}{\omega}$$

$$\omega_o = \sqrt{\frac{d_{xc}}{d_{xl}}} \qquad K = \frac{1}{2 \cdot (d_{xc} - d_{xl})} \qquad E = -K \cdot U_{di0} \cdot cos(\alpha) - \frac{I_d}{2}$$

$$\omega_n = \sqrt{\frac{1}{L_c \cdot C}} \qquad B = \frac{\sqrt{2} \cdot U_{c,LL} \cdot \omega \cdot cos(\alpha)}{2 \cdot L_c \cdot (\omega_n^2 - \omega^2)} \qquad D = \frac{-\sqrt{2} \cdot U_{c,LL} \cdot \omega \cdot sin(\alpha)}{2 \cdot L_c \cdot (\omega_n^2 - \omega^2)}$$

$$F = K \cdot \omega_o \cdot U_{di0} \cdot sin(\alpha) + \frac{\pi \cdot \omega_o}{3} \cdot I_d - \frac{3 \cdot \omega_o}{2 \cdot \pi \cdot d_{xc}} \cdot \Delta V_1$$

$$\lambda = atan2(D, B) \qquad \beta = atan2(F, E) \qquad Y = \sqrt{E^2 + F^2}$$

The average DC voltage is calculated as:

$$U_d = U_{di0} \cdot \frac{cos(\alpha + \mu) + cos(\alpha)}{2} + \left(1 - \frac{3}{4 \cdot \pi} \cdot \mu\right) \cdot (\Delta V_1 - \Delta V_2) \tag{20}$$

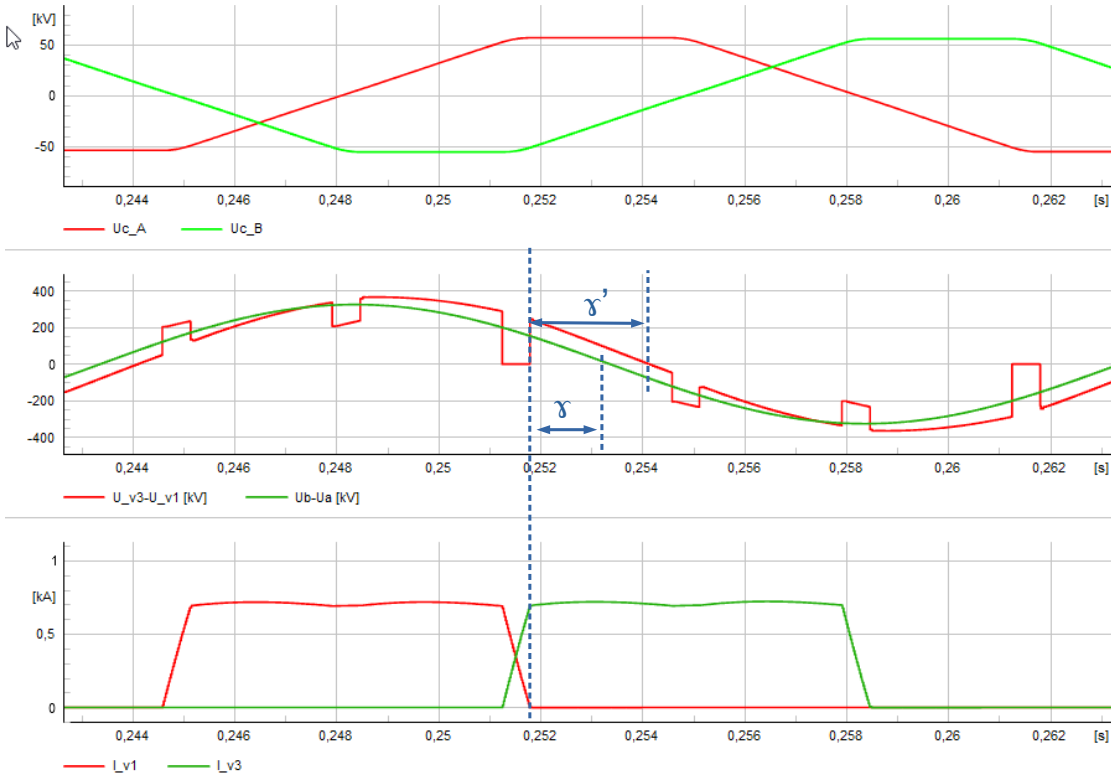


Figure 1.9: Capacitor voltages Uc_A and Uc_B, commutating voltage $U_b - U_a$, valve voltage $U_{v3} - U_{v1}$ and valve currents I_{v1} and I_{v3} for a CCC inverter.

and ΔV_1 and the overlap angle μ are calculated solving iteratively the following equations:

$$\Delta V_1 = \frac{2 \cdot \pi \cdot d_{xc}}{3 \cdot [1 + cos(\omega_o \cdot \mu)]} \cdot \Big[I_d \cdot \frac{\mu}{2} - K \cdot U_{di0} \cdot sin(\alpha + \mu) + \frac{1}{\omega_o} \cdot \left(K \cdot U_{di0} \cdot cos(\alpha) + \frac{I_d}{2} \right) \cdot sin(\omega_o \cdot \mu) + K \cdot U_{di0} \cdot sin(\alpha) \cdot cos(\omega_o \cdot \mu) + \frac{\pi}{3} \cdot I_d \cdot (cos(\omega_o \cdot \mu) - 1) \Big] \quad (21)$$

$$\frac{Id}{2} - |K \cdot U_{di0}| \cdot cos(\mu - \lambda) - Y \cdot cos(\omega_o \cdot \mu - \beta) = 0 \quad (22)$$

ΔV_2 is then calculated as:

$$\Delta V_2 = \frac{\pi}{3} \cdot I_d \cdot \mu \cdot d_{xc} - \Delta V_1 \quad (23)$$

If no internal transformer is defined, the line-line commutation voltage $U_{c,LL}$ is calculated as:

$$U_{c,LL} = \sqrt{3} \cdot \left| \frac{\underline{U}_{LL}}{\sqrt{3}} + j \cdot X_c \cdot \underline{I}_{L1} \right| \tag{24}$$

and if the internal transformer is defined:

$$U_{c,LL} = |\underline{U}_{LL}| \tag{25}$$

For unbalanced fundamental frequency calculations, $U_{c,LL}$ is calculated using only the positive sequence voltages and currents. U_{LL} is the line-line terminal voltage.

The reactive power is calculated as:

$$Q = \sqrt{3} \cdot \underline{U}_{c,LL} \cdot \underline{I}_1^* \tag{26}$$

where

$$\underline{I}_1 = \frac{\sqrt{6} \cdot e^{-j \cdot \alpha}}{\pi} \cdot \left[\frac{I_d}{2} \cdot (1 + e^{-j \cdot \mu}) + \frac{K \cdot U_{dio}}{2} \cdot \left[\frac{e^{-j \cdot \alpha}}{2} \cdot (1 - e^{-j \cdot 2\mu}) + j \cdot \mu \cdot e^{j \cdot \alpha} \right] + \frac{Y}{\omega_o^2 - 1} \cdot \left[e^{-j \cdot \mu} \cdot [cos(\omega_o \cdot \mu - \beta) + j\omega_o \cdot sin(\omega_o \cdot \mu - \beta)] - cos(\beta) + j\omega_o \cdot sin(\beta) \right] \right] \tag{27}$$

1.1.3 Commutation reactance

Specifying the correct values of the commutation reactances is important to get realistic values for the commutation (overlap) angles. The commutation reactances should be referred to the converter-side voltage of the transformer.

The commutation reactance is the reactance between the commutating voltage and the converter valves. The voltage where its waveform is reasonably sinusoidal (under normal operating conditions) can be taken as the commutating voltage. This is normally the case where filtering and phase-shifting is provided at the converter stations [1].

Typically, it is sufficient to take the voltage at the grid-side of the converter transformer as commutation voltage. Therefore, as commutation reactance, the leakage reactance of the transformer can be assumed (the transformer leakage reactance is usually the biggest part of the total reactance from the Grid side). The commutation reactance can be calculated as:

$$X_c = \frac{u_{kr}}{100} \cdot \frac{U_{r,conv}^2}{S_r} \quad [\Omega] \tag{28}$$

where:

- u_{kr} is the converter transformer short-circuit voltage in %,
- S_r is the rated power,
- $U_{r,conv}$ is the rated voltage on the converter side of the transformer. If a built-in transformer is used, the converter-side transformer voltage can be calculated using the parameters

$Unom$ and $tapnom$ defined in the type (it is here assumed that the nominal rated AC voltage $Unom$ is the same as the built-in transformer rated voltage on the grid side):

$$U_{r,conv} = Unom \cdot tapnom \tag{29}$$

If the rated DC voltage of the converter is known, the rated voltage of the converter-side of the transformer can be calculated using the following equation:

$$U_{r,conv} = \frac{\pi}{3 \cdot \sqrt{2}} \cdot \frac{U_d}{\cos(\alpha) - d_{xr}} \tag{30}$$

For more detailed studies, the presence of local plant (e.g. synchronous compensators and generators) needs to be taken into account for the commutation reactances since it has influence on the commutation process [1]. In [1], two practical examples for the calculation of the commutation reactances are presented. An external transformer needs to be modelled in this case.

1.1.4 Converter Transformer

The converter transformer can be modelled internally in the model or externally. The commutation reactance needs to be included in the voltage which is being considered for calculating the ideal no-load dc voltages of the converters. Depending if the transformer is modelled internally or externally, the voltage is calculated as follows:

- Internal (built-in) transformer

  The AC voltage drop due to the leakage (commutating) reactance is already considered in the model and only the tap position, nominal ratio and phase shift need to be taken into account:

  $$\underline{U}_{ac,conv} = \underline{U}_{term} \cdot tapnom \cdot nntap \cdot e^{j \cdot nt2ag \cdot \frac{\pi}{6}} \tag{31}$$

  where *nt2ag* is the built-in transformer phase shift and $\underline{U}_{term}$ is the voltage where the element is connected (grid terminal).

  The built-in transformer has a tap-changer on the LV side to control the converter-side voltage. The tap changer setting is specified with the actual winding ratio *nntap* parameter, which must be between the maximum and minimum turns-ratio specified in the type. The actual winding ratio *nntap* is given in per unit of the nominal turns-ratio *tapnom* of the converter transformer, also specified in the type.

  The built-in transformer can be configured as:

  - Fixed Tap
  - alpha setpoint control: control of firing angle to specified value
  - gamma setpoint control: control of extinction angle (LCC) or commutation margin (CCC) to specified value
  - alpha range control: control of firing angle between min/max range
  - gamma range control: control of extinction angle (LCC) or commutation margin (CCC) between min/max range
  - Vdc setpoint control: control of DC-voltage to specified value

  In the *alpha setpoint control*, *gamma setpoint control* and *Vdc setpoint control* cases, the *Relaxation Factor* (*krelax*) parameter can be entered to speed up (set a factor of greater than 1.0) or slow down (set a factor of less than 1.0) the tap controller action.

- External transformer

  If an external transformer is used for the converter model, the AC voltage drop over the transformer is calculated using the commutation reactance specified in the converter dialogue window:

$$\underline{U}_{ac,conv} = \underline{U}_{term} + \jmath \cdot X_c \cdot \underline{I} \tag{32}$$

  where $\underline{I}$ is the current flowing through the element,and X_c is the commutation reactance.

The rated AC current on the converter side of the transformer equals the RMS value of the total AC current which, neglecting commutation effects, consists of rectangular pulses with amplitude equal to the rated DC current I_d and duration of 120°. The rated AC current on the LCC converter side of the transformer is found as:

$$I_{r,conv} = \sqrt{\frac{2}{3}} \cdot I_d \tag{33}$$

Note that this current is the RMS value of the total AC current, and not only of its fundamental frequency component, which is calculated instead as in equation 16. Hence the rated power for a LCC converter transformer can be calculated as

$$S_r = \sqrt{3} \cdot I_{r,conv} \cdot U_{r,conv} = \sqrt{2} \cdot I_d \cdot U_{r,conv} = \frac{\pi}{3} \cdot \frac{P_d}{cos(\alpha) - d_{xr}} \tag{34}$$

1.1.5 Losses

The model of the HVDC does not include the effects of losses so far. The losses in the converter bridge are caused due to the different components, i.e. the resistances of valves, transformers, smoothing reactances.

An exact representation of the losses associated with the converter station is very sophisticated, so it is common practice to model the losses in the Load Flow analysis as an equivalent series resistance on the DC side. Two more terms accounting respectively for the forward voltage drop in the thyristors and no-load losses depending on the DC voltage are also considered. In *PowerFactory*, rectifier/inverter losses are specified in fundamental frequency models as:

- No-load losses: specified with the input parameter $Pnold$ in [kW].
- Forward voltage drop losses: specified with the input parameter $swtLossFactor$ in [KW/A].
- Resistive losses: specified with the input parameter $resLossFactor$ in [Ohm].

The converter losses in [MW] are calculated as:

$$DCPoleLosses = G_{noload} \cdot U_d^2 + resLossFactor \cdot I_d^2 + V_{drop} \cdot I_d \tag{35}$$

with:

- $G_{noload} = \frac{P_{nold}}{1000 \cdot U_{nom,DC}^2}$ where U_{DCnom} is expressed in kV
- $V_{drop} = sign(I_d) \cdot swtLossFactor \cdot (1 - exp^{-200 \cdot |I_d|})$

To take into account the losses, equation 7 must be modified, both for LCC rectifier and inverter, as:

$$U_d = U_{d0} \cdot \cos(\alpha) + \frac{3}{\pi} X_c \cdot I_d + resLossFactor \cdot I_d + V_{drop} \tag{36}$$

For a CCC, losses are accounted for by modifying equation 20 as:

$$U_d = U_{di0} \cdot \frac{cos(\alpha + \mu) + cos(\alpha)}{2} + \left(1 - \frac{3}{4 \cdot \pi} \cdot \mu\right) \cdot (\Delta V_1 - \Delta V_2) + resLossFactor \cdot I_d + V_{drop} \tag{37}$$

The representation of station losses of a rectifier for load flow calculations is shown in Figure 1.10. Note in the figure and in the equation above, that the DC current is negative for the rectifier operation due to the representation with load-orientation.

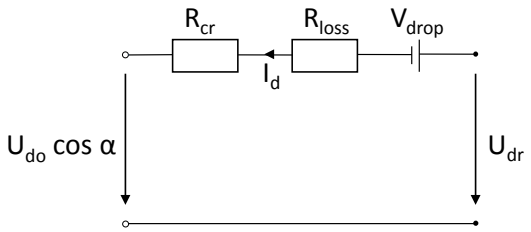


Figure 1.10: Simplified modelling of losses for a 6-pulse LCC converter

The total losses for the element in MW and Mvar are obtained as:

$$\begin{aligned} Ploss &= P_{busac} + P_{busdc} = DCPoleLosses \\ Qloss &= 0 \end{aligned} \tag{38}$$

where P_{busac} and P_{busdc} are calculation parameters available for the element.

The load losses are calculated as the difference between the total losses and the no-load losses:

$$\begin{aligned} Plossld &= Ploss - Plossnld \\ Qlossld &= 0 \end{aligned} \tag{39}$$

where the no-load losses are $Plossnld = G_{noload} \cdot U_d^2$.

1.1.6 Unbalanced Operation of LCC

When the network voltages are unsymmetrical, the periods for natural conduction of the valves are not the same and the DC voltage will in general be made of six pulses with different duration and amplitude. The ideal no-load DC voltage can be calculated by taking the average of the pulses over half a period. As an example, with reference to Figure 1.11, at $\omega t = \theta_{ca} + \pi$ the DC current starts flowing in phase *a* and returns back through phase *b* until $\omega t = \theta_{bc}$.

For a thyristor converter, these angles can be delayed by α_a and α_c respectively. During this period, the DC voltage is equal to the line-line voltage U_{ab}. In *PowerFactory*, the thyristor converter implements the Equidistant Pulse Control, igniting the valves at equal time intervals [2].

The angles θ_{ab}, θ_{bc} and θ_{ca}, at which the corresponding line-line voltages cross zero and become positive, are calculated internally in the model, given the terminal voltage phasors. This holds only for load flow and RMS simulations.

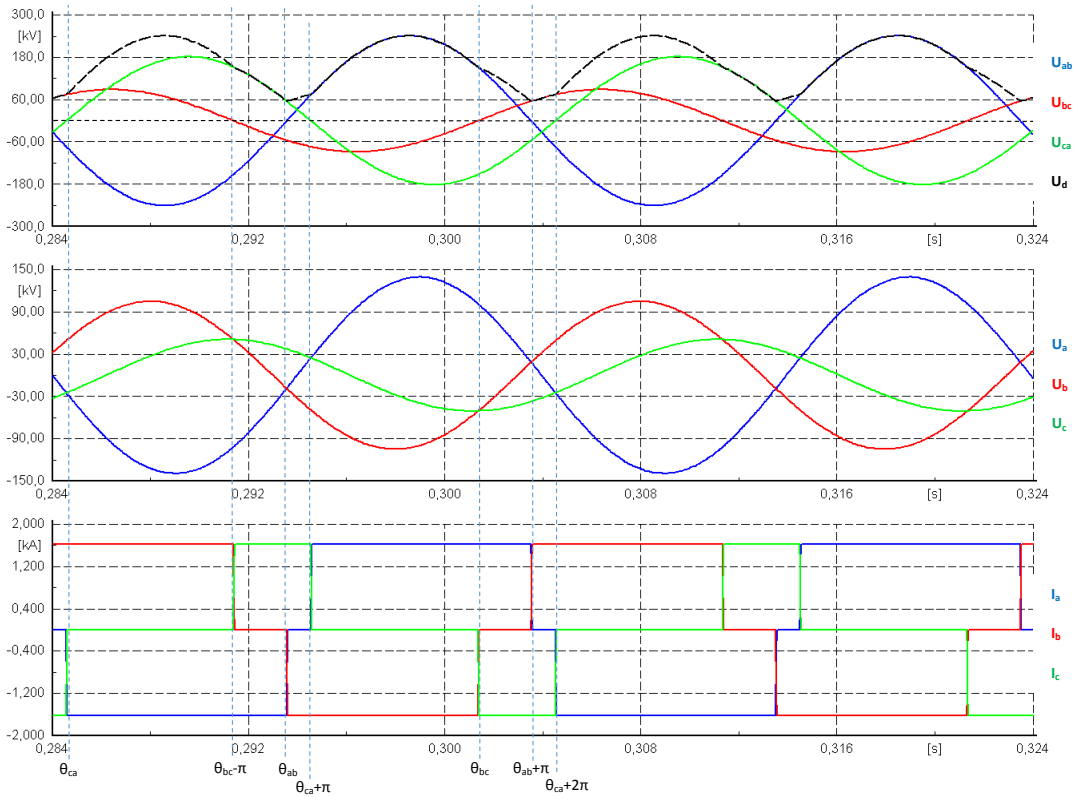


Figure 1.11: Currents and voltages for diode bridge rectifier with unbalanced network voltages and smoothing reactor on the DC side

By taking the average of the resulting three pulses over half a period, the ideal no-load DC voltage is calculated in load flow and RMS simulations as:

$$U_d = \frac{1}{\pi} \cdot \{U_{ab} \cdot [\cos(\theta_{ca} + \pi + \alpha_a - \theta_{ab}) - \cos(\theta_{bc} + \alpha_c - \theta_{ab})] + U_{bc} \cdot [\cos(\theta_{ab} + \pi + \alpha_b - \theta_{bc}) - \cos(\theta_{ca} + \alpha_a - \theta_{bc})] + U_{ca} \cdot [\cos(\theta_{bc} + \pi + \alpha_c - \theta_{ca}) - \cos(\theta_{ab} + \alpha_b - \theta_{ca})]\} \quad (40)$$

In unsymmetrical load flow and RMS simulations, a PLL will measure the angle θ_1 of the positive zero-crossing of the positive sequence line-line voltage. The firing angles of the valves are then calculated in order to obtain an interval between the firing pulses of 60°. Therefore, the firing angles α_a, α_b and α_c will not be identical but differ according to the phase-shift of the phase voltages. The firing angles of the three phases are calculated internally in the model as (the negative sign is due to the fact that negative angles imply a lagging phase in equation 40):

$$\alpha_a = -[\alpha + (\theta_{ca} - (\theta_1 - 4\pi/3))] \quad (41)$$

$$\alpha_b = -[\alpha + (\theta_{ab} - \theta_1)] \quad (42)$$

$$\alpha_c = -[\alpha + (\theta_{bc} - (\theta_1 - 2\pi/3))] \quad (43)$$

In general it is not possible to obtain a 60° interval between the firing pulses, since this would require leading firing angles. As a consequence, for very large unbalances the pulses will still have different length even after applying different firing pulses for each phase.

The AC currents are affected by the different length of the pulses. Assuming a constant DC current, the amplitude of the fundamental frequency component of the phase currents is in general no longer equal because the conduction periods for each phase are different, as can be seen in Figure 1.11. Let θ_{Ii} represent the conducting time of phase *i*. The RMS value of the fundamental frequency component of each phase current is easily calculated by Fourier series expansion as:

$$I_{L1i} = k \cdot \frac{4}{\pi} \cdot \frac{I_d}{\sqrt{2}} \cdot \sin(\frac{\theta_{Ii}}{2}) \tag{44}$$

The DC power is then calculated as the sum of the real power of all phases on the secondary side of the transformer:

$$P_{dc} = P_a + P_b + P_c \tag{45}$$

1.1.7 Unbalanced Operation of CCC

Under unbalanced conditions in the grid the CCC will not inject any negative and zero sequence current. The CCC model is based solely on the positive sequence, according to the equations presented in section 1.1.2.

2 Load Flow Analysis

In load flow analysis, it is common practice not to specify control variables directly but to define the controlled variables instead. The control variable (the firing angle α) is then resulting from the Load Flow calculation.

In the Load Flow command, several common control characteristics are supported by the HVDC converter model. Meaning and typical application of the various control modes are the following:

- Vdc: The firing angle is adjusted to obtain a predefined value for the DC voltage of the converter. This control mode is typically used at the inverter side of an HVDC transmission system.
- Vac: Specifies the magnitude of the AC voltage at the converter terminals, when the DC voltage is controlled externally. No typical application.
- Pdc: The transmitted DC power is held constant. Typically used for rectifier side in HVDC systems.
- P: The transmitted AC power is held constant. Typically used for rectifier side in HVDC systems.
- Q: Specifies the amount of reactive power absorbed by the converter. No typical application.
- I: The DC current of the converter is held constant. Typically used for rectifier control of an HVDC transmission system.
- Gamma: The extinction angle γ (LCC) or commutation margin γ' is specified. Normally the inverter side of an HVDC system is controlled to a minimum γ (γ').
- EXT: The firing angle α is specified as an input to the model, provided by a *Controller* which must be specified with the parameter *pctrl*. The *Controller* is a line-commutated rectifier/inverter element (*ElmRec* or *ElmRecmono*) as well. The *EXT* control mode is useful in a 12-pulse arrangement with two converters: one converter is the *Controller* performing one of the other control modes, while the second converter is in *EXT* control mode.

These control modes are enabled if the flag *Automatic Firing Angle Control* is selected. Otherwise, the firing angle α is set equal to the *Actual Firing Angle*, specified with the parameter *alpha_set*.

Minimum and maximum firing angle limits can be specified. If the converter reaches one of these limits, the firing angle will remain constant at the limit and the converter cannot perform the chosen control function.

A minimum value of the extinction angle, *gammamin*, can also be entered. The angle *gammamin* represents a safe value of γ in order to avoid commutation failure in normal operating conditions; it does not represent the limiting value for commutation failure. If the inverter extinction angle γ reaches this limit, a warning is printed in the output window. For inverters in Vdc (EXT) control mode, the option *Consider minimum extinction angle (gammamin) for control* is available. If the option is selected, the inverter will no longer control the DC side voltage but will switch to a gamma-control mode with *gammamin* as setpoint if gammamin is reached. For a CCC converter a minimum value of the commutation margin *gammaPrimemin* is specified.

In load flow calculations, commutation failure for an LCC is assumed to take place only when the sum of the firing angle α and of the overlap angle μ would be higher than 180° (negative

γ). A warning for commutation failure is printed in the output window. In this case, the load flow is still calculated assuming the overlap angle equal to zero. However, this may not represent a feasible operating point. Commutation failure for a CCC is assumed to take place when the commutation margin is negative.

During load flow calculation, if the thyristor converter current is very low, the converter current is set to zero. The thyristor converter voltage is set to zero with the firing angle equal to 90°. A message is sent to the output window warning about zero current flowing in the converter.

Furthermore the control of the tap-changers of the converter transformer can be chosen between:

- Fixed Tap: The position of the tap-changers is fixed to a given winding ratio.
- alpha setpoint control: The secondary voltage is adjusted by the tap-changers to obtain a specified setpoint of the firing angle. This is typically used at the rectifier station of the HVDC.
- gamma setpoint control: The tap-changers are controlled to obtain a specified setpoint of the extinction angle (LCC) or commutation margin (CCC). This is typically used at the inverter station of the HVDC.
- alpha range control: The secondary voltage is adjusted by the tap-changers to control the firing angle between *alphamin* and *alphamax* range. This is typically used at the rectifier station of the HVDC.
- gamma range control: The secondary voltage is adjusted by the tap-changer to control the extinction angle (LCC) between *gammamin* and *gammamax* range. For a CCC converter the tap-changer controls the commutation margin between *gammaPrimemin* and *gammaPrimemax* range. This is typically used at the inverter station of the HVDC.
- Vdc setpoint control: the tap is adjusted to control the DC-voltage to a specified value. This can be used at the inverter station of a CCC HVDC system while the firing angle controls the AC voltage.

Besides the firing angle control modes, the load flow page of the converter also comprises additional information for the converter transformer. Here the commutation reactance X_c is specified as the leakage reactance of the transformer, which is important for the calculation of the commutation angle. Also the phase-shift of the converter transformer can be entered here. This information is needed, when designing 12-pulse thyristor bridges with 30° phase-shift between two converters in series to reduce harmonic currents fed into the network.

Note: This information is also needed, when no built-in transformer is selected in the converter type! The value of the commutation reactance is specified as the reactance of the converter transformer, modelled externally. The value of the commutation reactance is used to estimate the voltage on the transformer primary side, given the converter terminal voltage and current. The estimated voltage is used to calculate the ideal no-load DC voltage. Specifying the correct value of the commutation reactance is also important to get realistic values for the commutation angle.

Note: It is advised not to connect the converter to a DC terminal (possibly grounded) with extremely low nominal voltage, e.g. $<$ 1 V. This may result in numerical issues possibly causing wrong results.

2.1 P-setpoint Adaption

When the converter is in *P* or *Pdc* control mode, usually the rectifier side in an HVDC system, the active power setpoint can be modified by the following controllers if selected:

- *Angle-difference dependent P-droop*
- *Active power participation*

When the ***Angle-difference dependent P-droop*** option is selected, the active power setpoint is modified. If the option *Consider active power setpoint* is selected, the new active power setpoint for the rectifier and inverter case is given according to:

$$P_r = P_{r,set} + Kpphi \cdot (phiu_{local} - phiu_{remote})$$
$$P_i = P_{i,set} - Kpphi \cdot (phiu_{local} - phiu_{remote})$$

If the option *Consider active power setpoint* is not selected, the new active power setpoint for the rectifier and inverter case is given according to:

$$P_r = +Kpphi \cdot (phiu_{local} - phiu_{remote})$$
$$P_i = -Kpphi \cdot (phiu_{local} - phiu_{remote})$$

When the ***Active power participation*** option is selected, the active power setpoint is modified. If the option *Consider active power setpoint* is selected, the new active power setpoint for the rectifier and inverter case is given according to:

$$P_r = P_{r,set} + K_{part} \cdot P_{meas}$$
$$P_i = P_{i,set} - K_{part} \cdot P_{meas}$$

If the option *Consider active power setpoint* is not selected, the new active power setpoint for the rectifier and inverter case is given according to:

$$P_r = +K_{part} \cdot P_{meas}$$
$$P_i = -K_{part} \cdot P_{meas}$$

where:

- $P_{r,set}$ is the active power setpoint, rectifier case.
- $P_{i,set}$ is the active power setpoint, inverter case.
- P_r is the modified active power setpoint, rectifier case.
- P_i is the modified active power setpoint, inverter case.
- $Kpphi$ is the specified factor for *Angle-difference dependent P-droop*.

- K_{part} is the specified factor for *Active power participation*.
- $phiu_{remote}$ is the positive-sequence voltage angle of the remote busbar.
- $phiu_{local}$ is the positive-sequence voltage angle of the local busbar.
- P_{meas} is the active power measured (assumed positive with load orientation) at a specified cubicle/boundary.

The *Angle-difference dependent P-droop* and *Active power participation* options can be used to adapt the active power of the converter depending on the active power flow on a parallel AC line. When the *Active power participation* option is selected, the sign of the parameter *Kpart* depends on the orientation of the power flow at the point where the parameter *Pmeas* is measured. Figure 2.1 shows how to correctly define the sign of *Kpart*. In the example, the converter INV is performing the active power control and has the *Active power participation* option selected.

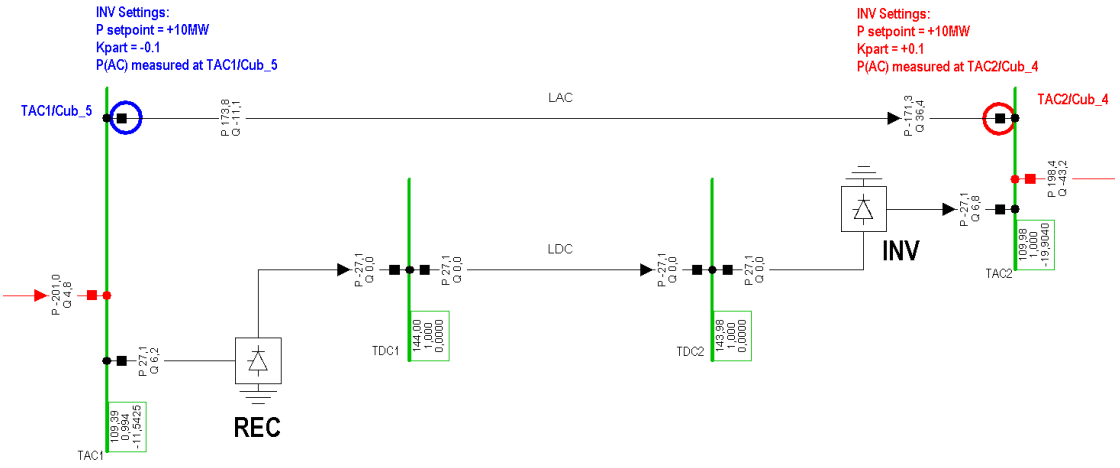


Figure 2.1: *Active power participation* example for converter INV. Active power setpoint is P=10MW.

2.2 Active Power and DC Voltage Input

It is possible to control the active power injection and the DC voltage of the converter through an external *QDSL* model with the input *pset* and *udcset*. The *Automatic Firing Angle Control* or the *HVDC LCC Control* must be activated.

To control the active power with the input *pset* the *Control-Characteristic* must be set to ”Pdc” or ”P”.

To control the DC voltage with the input *udcset* the *Control-Characteristic* must be set to ”Vdc”.

It is not possible to reverse the power power flow with a *QDSL* model as compared to the original setting in the element.

2.3 HVDC LCC system

If a HVDC LCC system has been defined, the LoadFlow model is equivalent to the LoadFlow model used by the ElmHvdclcc element. Please refer to the Technical Reference of the ElmHvdclcc for more information. This control mode is also available for the CCC converter.

3 Short-Circuit Analysis

3.1 VDE/IEC Short-Circuit

By default, rectifiers and inverters are not considered in any VDE/IEC short-circuit calculation. With the option *Static converter-fed drive* enabled, the element can be used to represent reversible static converter-fed drives. In this configuration, the equivalent circuit shown in Figure 3.1 applies.

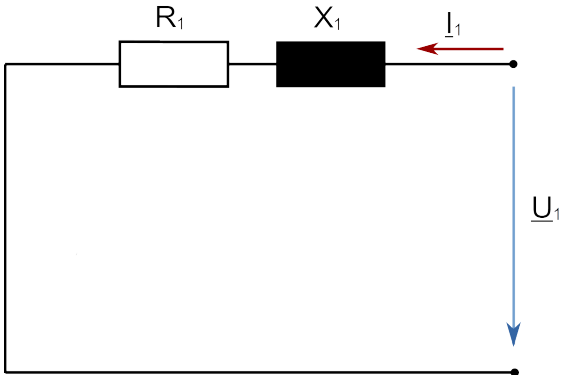


Figure 3.1: Equivalent positive sequence circuit for use in the VDE/IEC short-circuit calculation

The impedance $\underline{Z}_1 = R_1 + j \cdot X_1$ is calculated as follows:

$$X_1 = \frac{1}{3 \cdot \sqrt{1 + 0.1^2}} \cdot \frac{{U_N}^2}{P_N} \tag{46}$$

$$R_1 = 0.1 \cdot X_1 \tag{47}$$

where:

- U_N is the rated AC voltage in kV (*t:Unom*)
- P_N is the rated active power in MW (*t:Pnom*)

Steady-State and Breaking Current

As per IEC60909, static converter-fed drives do not contribute to breaking or steady-state currents.

Unbalanced Faults

As per IEC60909, static converter-fed drives are always neglected for unbalanced faults.

3.2 Complete Short-Circuit

Rectifiers and inverters will retain the current obtained from the pre-fault load flow during complete short-circuit calculations. If the complete short-circuit does not use load flow initialisation rectifiers and inverters are neglected.

The option *Consider undervoltage blocking* allows for integrated protective controls to be considered automatically during the short-circuit calculation.

If the threshold is enabled, the short-circuit calculation is carried out as follows:

1. Calculate the subtransient short-circuit;
2. Evaluate the AC side voltage against the threshold and activate the blocking if necessary;
3. If the blocking is activated, the subtransient short-circuit is re-calculated.

The state of the blocking determined for the subtransient calculation is retained for the transient calculation.

For unbalanced short-circuit calculations, it is sufficient for one phase to fall below the threshold to trigger the blocking.

3.3 ANSI Short-Circuit

Rectifiers are not considered in any ANSI short-circuit calculation. Inverters are considered if the option *Short-Circuit Contribution* is enabled.

Inverters are considered with the equivalent circuit shown in Figure 3.1 for the calculation of momentary current contributions. Inverters are neglected for the calculation of interrupting and 30-cycle current contributions.

Unbalanced Faults

For unbalanced faults, $\underline{Z}_2 = \underline{Z}_1$ and $\underline{Z}_0 = \infty$ are assumed.

3.4 IEC 61363

Rectifiers and Inverters are not considered in the IEC 61363 short-circuit calculation.

3.5 DC Short-Circuit

Only the *ElmRecmono* is considered in the calculation of DC short-circuits according to IEC 61660 and ANSI/IEEE 946. Additional parameters required to perform DC short-circuit calculations according to the standards can be entered in the *DC Short-Circuit* page of the element *ElmRecmono* and of the type *TypRec*. In these pages, data about the commutation resistance, AC side impedance, DC side resistance and inductance, converter connection type and voltage factor are specified.

4 Harmonics

The currents of the 6-pulse thyristor-controlled converter, which are shown in Figure 1.5 with the commutation effect neglected and in Figure 1.7 including commutation, are characteristic waveforms. From these curves it can easily be seen, that the currents not only have a large 50-Hz-component but also cause a flow of harmonic currents of higher orders. Hence the most accurate harmonic model of the HVDC converter is a harmonic current source. The order of the harmonic currents is calculated as

$$h = 6 \cdot n \pm 1 \tag{48}$$

where $n = 1,2...$ is an integer value.

The injected harmonic current component can be calculated as:

$$I_h = \frac{I_{L1}}{h} \tag{49}$$

The polarity of the harmonics (angle) is 180° for the 5th, 11th, etc. harmonics (represented in the negative sequence) and 0° for the 7th, 13th, etc. harmonics (represented in the positive sequence).

The harmonic spectrum can be defined using three different models:

- *Idealised 6-pulse-bridge*
  The typical spectrum of the converter is assumed up to a specified number (normally 31) using the above two equations with the commutation reactance neglected. This assumption usually causes the harmonic currents to be larger than in reality, but gives a good approximation.
- *Consider overlap angle*
  In this model the harmonic currents are obtained same as in the *Idealised 6-pulse-bridge* model and then are multiplied with a reduction factor calculated using the overlap angle. In this case, under typical loading conditions the 11th and 13th harmonics are 30% to 40% of the harmonics calculated using the *Idealised 6-pulse-bridge* model [2].
- *User defined*
  To represent the converter in a more realistic way, a harmonic current source can be defined and the amplitude and angle of the harmonic currents can be defined as shown in Figure 4.1. A balanced and unbalanced representation can be chosen. More information can be derived from the Technical Reference of the type “Harmonic Sources” (*TypHmccur*).

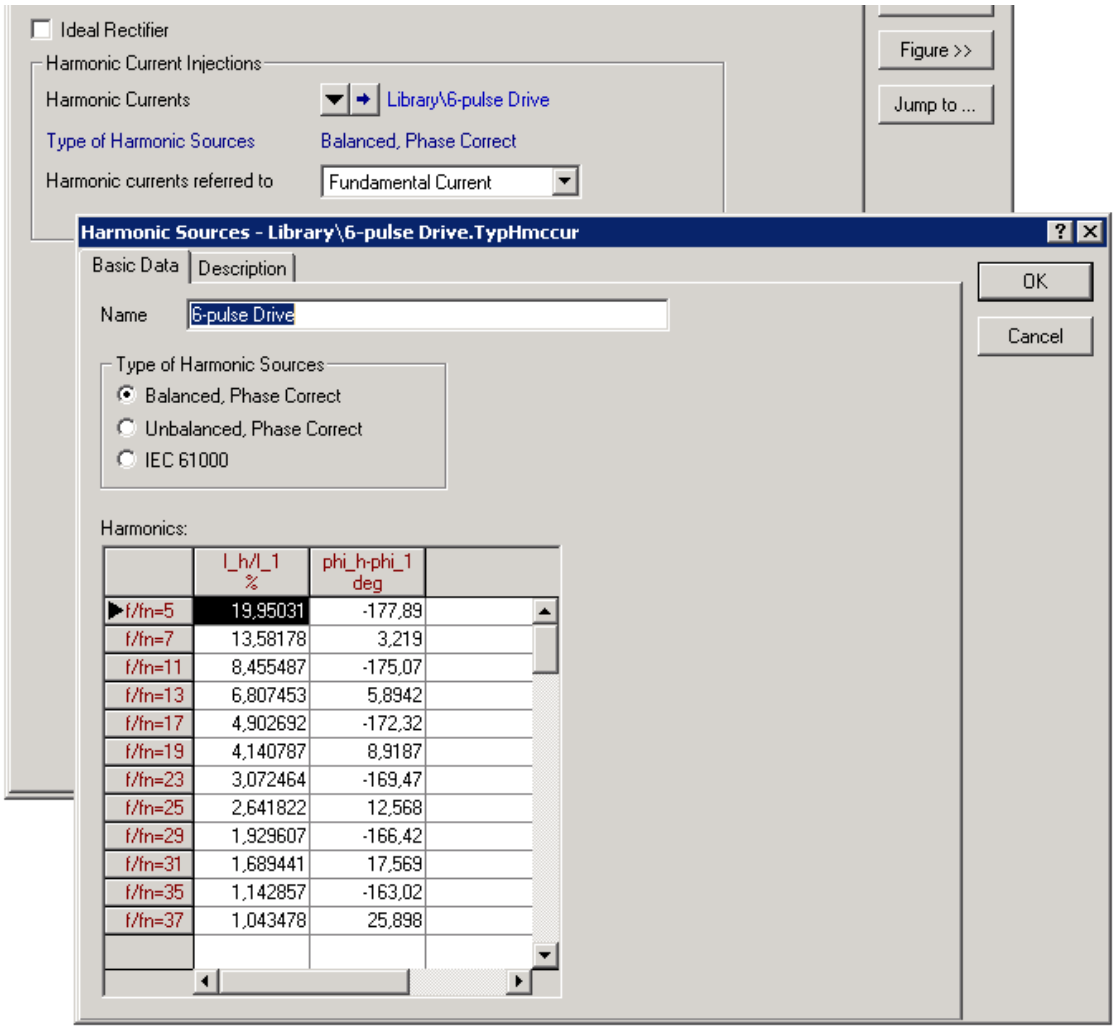


Figure 4.1: Example for a converter representation as harmonic current source

5 Dynamic Simulation

5.1 RMS Simulation

The stability model uses the same equations as described in section 2 (Load Flow analysis).

The converter transformer data, commutation reactance and phase shift, are identical with the values specified on the load flow page of the element.

A separate minimum extinction angle for commutation failure *gammamindyn* can be specified in the *RMS* page. This angle specifies the minimum extinction angle below which commutation failure is assumed to take place. When the extinction angle γ reaches the specified *gammamindyn*, a message warning for commutation failure is printed in the output window, the DC side is short-circuited and the converter current is assumed equal to zero. Notice that the *gammamindyn* angle has a different meaning than the *gammamin* angle specified for control purposes in the load flow page. For a CCC converter a separate minimum commutation margin for commutation failure *gammaPrimemindyn* can be specified.

With the signal *short_dc* is possible to short-circuit the DC side of the element, in order to bypass the valves. The DC voltage and the AC current go to zero during active *short_dc*. With the signal *block_all* all thyristors will be blocked. The AC and DC current are then both zero.

It is possible to select the way the extinction angle γ is handled when the converter is blocked through the parameter *gammaMode*: The angle can either be set to zero or kept constant. An additional output signal *gamma_state* is available. The value of *gamma_state* is 0 in normal operation, 1 if the converter is blocked (AC side), 2 if the converter is shorted (DC side) or if in commutation failure. For a CCC converter it is possible to select the way the commutation margin γ' is handled when the converter is blocked through the parameter *gammaPrimeMode*.

The zero sequence current is always zero in RMS simulations.

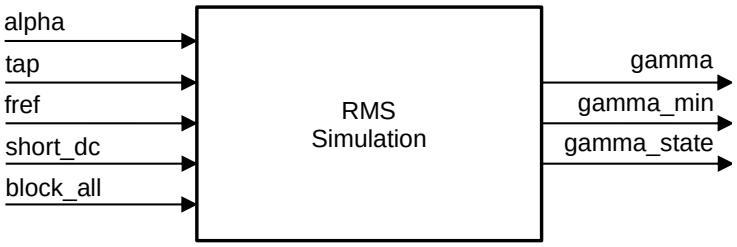


Figure 5.1: Input/Output definition of the HVDC converter model for stability analysis (RMS-simulation)

Note: It is advised not to connect the converter to a DC terminal (possibly grounded) with extremely low nominal voltage, e.g. < 1 V. This may result in numerical issues possibly causing wrong results.

5.1.1 High DC current mode, LCC

The high DC current mode can be optionally selected on the element RMS page for an LCC. In *PowerFactory* versions previous to 2022, the high DC current mode was always modelled. This option is not available for a CCC.

In RMS simulations, the behaviour of an LCC during low DC voltage / high DC current conditions is represented in *PowerFactory* by using a different model than the one used during normal operation and presented in section 1.1.1. This allows a more accurate prediction of the DC current during faulted conditions. The condition for switching between the normal operating condition and the high DC current mode is specified in this section.

The equivalent circuit of the converter when in high DC current mode is shown in Figure 5.2.

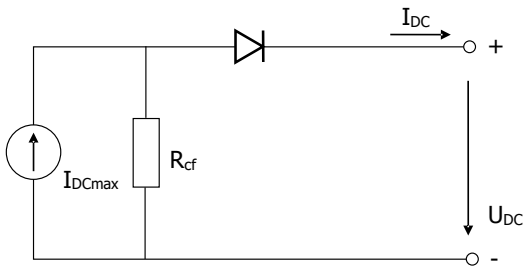


Figure 5.2: Equivalent DC circuit diagram, high DC current mode

The DC side of the LCC is represented through a Norton equivalent, with:

$$R_{cf} = 2 \cdot \sqrt{2} \cdot \frac{3}{\pi} \cdot X_c \tag{50}$$

$$I_{DCmax} = \frac{\sqrt{2} \cdot |\underline{U}_{c,LL}|}{\sqrt{3} \cdot X_c} \tag{51}$$

and the line-line commutating AC voltage is calculated as:

$$\underline{U}_{c,LL} = \sqrt{3} \cdot (\underline{U}_{LL}/\sqrt{3} + j \cdot X_c \cdot \underline{I}_{L1}) \tag{52}$$

where X_c is the commutation reactance, $\underline{U}_{LL}$ is the terminal AC line-line voltage and $\underline{I}_{L1}$ is the fundamental frequency component of the AC current.

The DC-current is thus given as:

$$I_{DC} = I_{DCmax} - \frac{U_{DC}}{R_{cf}} \tag{53}$$

The transition from normal operation to high DC current mode is done if the DC current calculated assuming the model presented in section 1.1.1 is higher than the DC current calculated for the high current diode mode according to Equation 54.

5.2 EMT Simulation

For the electro-magnetic transient simulation the detailed modelling of all six thyristors or diodes is necessary. Here the converters are modelled as shown in Figure 1.2. This detailed model represents the discrete valves including on- and off-resistances of the switches (R_{on}, G_{off}) and the elements of the snubber-circuits in parallel (shown in Figure 5.3). These values can be defined in the *EMT* page of the type *TypRec*.

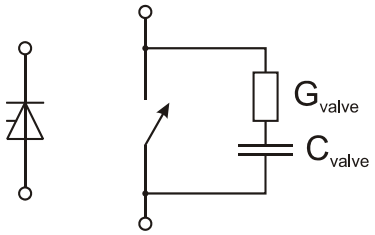


Figure 5.3: Detailed valve representation for EMT-simulations

For triggering the valves the built-in trigger-circuit can be used, which converts the firing angle supplied by the converter controller to the six correct firing signals of the discrete thyristors.

For an exact triggering of the valves, the times of zero-crossing of the AC voltages have to be measured. Since the 6-pulse converter does not have a built-in phase measurement, a PLL element (*ElmPhi__pll*) has to be provided. This means the output of a PLL *Fmeas* is required by the converter for accurate operation.

With the signal *short_dc* is possible to short-circuit the DC side of the element, through a valve with resistance equal to R_{on}. With the signal *block_all* all thyristors will be blocked. The AC and DC current are then both zero.

The converter transformer data, commutation reactance and phase shift, are identical with the values specified on the load flow page of the element. The minimum extinction angle *gamma-mindyn* is not used in EMT simulations.

If the built in transformer is used the zero sequence current is always zero.

Note: Modelling the converter transformer externally (i.e. the flag *Built-In Transformer* in *TypRec* is not selected) can cause problems due to the exact value of commutation reactance specified on the load flow page and the additional reactance inserted by the transformer element. Small errors for the commutation reactance value affect the commutation angle and the EMT simulation may not calculate the right initial conditions with sufficient accuracy. Hence, it is recommended to accurately define the commutation reactance if the transformer is modelled externally. Using the built-in transformer of the converter element instead (whenever possible), allows to get correct results in the EMT-simulation.

Note: If the converter is initially connected, the voltage angle is correctly initialized from the load flow solution. In case the converter is initially disconnected, the voltage angle is initialized to zero. To properly account for the angle shift of the external transformer, a parameter event must be defined for the signal *s:phiref* of each single converter before it is connected to the grid during EMT simulation. The value of must be set according to:

$$phiref = \frac{\pi}{3} - PhaseShift_rad \tag{54}$$

where

$$PhaseShift_rad$$

is the external transformer phase shift (due to its vector group). For ex., for a Yd1 transformer, set a value of PI/3-1*PI/6 = PI/6. For a Yy0, set a value of 0.

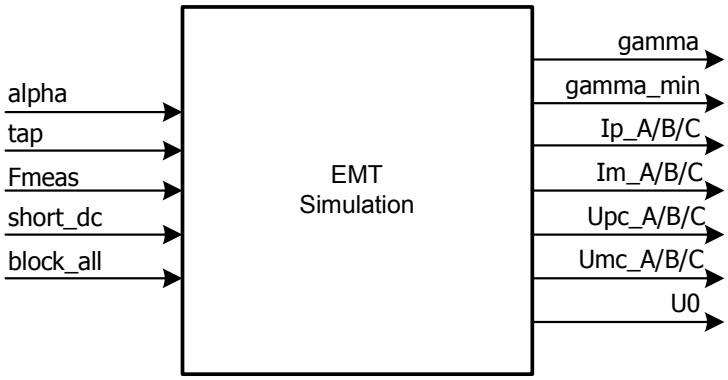


Figure 5.4: Input/Output definition of the HVDC converter model for stability analysis (EMT-simulation)

A Parameter Definitions

Table A.1: Parameters of the HVDC converter element

Parameter	Description	Unit
loc_name	Name	
typ_id	Type (*Typrec*)	
busac	Terminal AC (StaCubic)	
busac_bar	Terminal AC	
busdp	Terminal DC+ (StaCubic)	
busdm	Terminal DC- (StaCubic)	
busdc	Terminal DC (StaCubic)	
busdc_bar	Terminal DC	
outserv	Out of Service	
mode	Orientation (Rectifier/Inverter)	
bstp	Firing Angle (alpha-)Control: Control-Characteristic	
uset	Firing Angle (alpha-)Control: Voltage Setpoint	p.u.
Pset	Firing Angle (alpha-)Control: Power-Setpoint	MW
Qset	Firing Angle (alpha-)Control: Reactive Power-Setpoint	Mvar
Iset	Firing Angle (alpha-)Control: Current Setpoint	kA
gamma_set	Firing Angle (alpha-)Control: Extinction Angle (gamma) Setpoint	deg
pctrl	Firing Angle (alpha-)Control: Controller	
alphacn	Firing Angle (alpha-)Control: Automatic Firing Angle Control	
alpha_set	Firing Angle (alpha-)Control: Actual Firing-Angle	deg
alphamin	Firing Angle (alpha-)Control: Minimum Firing Angle	deg
alphamax	Firing Angle (alpha-)Control: Maximum Firing Angle	deg
gammamin	Firing Angle (alpha-)Control: Minimum Extinction Angle	deg
gammaminCtrl	Consider minimum extinction angle (gammamin) control	
ntrcn	Converter Transformer: Tap-Changer	
nntap	Converter Transformer: Actual Winding Ratio	p.u.
Xd [1]	Converter Transformer: Commutation Reactance	Ohm
nt2ag	Converter Transformer: Phase Shift	*30deg
iPphidrp	Angle difference dependent P-Droop	
Kpphi	Kpphi	MW/degree
p_b1phiu	Remote AC busbar (ElmTerm*)	
p_b2phiu	Local AC busbar (ElmTerm*)	
iPpart	Active power participation	
Kpart	Participation factor	
p_pmeas	P(AC) measured at (StaCubic*,ElmBoundary)	
iconfed	Static converter-fed drive	
i_int	Ideal Rectifier	
maxorder	Maximum Harmonic Order	
phmc	Harmonic Currents (TypHmccur)	
cTypHmc	Type of Harmonic Sources	
icurref	Harmonic Current Injections referred to	
Inom	Rated Harmonic Current Injection	kA
iAstabint	A-stable integration algorithm	
gammamindyn	Min. extinction angle for commutation failure	
gammaMode	Handling of extinction angle if rectifier is blocked	
comres	Commutation resistance (Only ElmRecmono)	Ohm
Racmax	Max. values: AC resistance (Only ElmRecmono)	Ohm

[1] The parameter Xd corresponds to the commutation reactance Xc defined in equation 28

Xacmax	Max. values: AC reactance (Only ElmRecmono)	Ohm
Racmin	Min. values: AC resistance (Only ElmRecmono)	Ohm
Xacmin	Min. values: AC reactance (Only ElmRecmono)	Ohm
volfac	IEC parameters: Voltage factor, c (Only ElmRecmono)	
gammaPrimeset	Commutation margin setpoint	deg
gammaPrimemin	Minimum commutation margin	deg
gammaPrimemax	Maximum commutation margin	deg
gammaPrimeminCtrl	Consider minimum commutation margin (gammaPrimemin)	
gammaPrimeMode	Handling of commutation margin (gamma_prime) during commutation failure	
gammaPrimemindyn	Min. commutation margin for commutation failure	deg
udctapset	Vdc setpoint	p.u.
iHighDCCur	Consider "High DC current mode"	

Table A.2: Parameters of the HVDC Converter Type

Parameter	**Description**	**Unit**
loc_name	Name	
Unom	Rated AC Voltage	kV
Unomdc	Rated DC-Voltage (DC)	kV
Pnom	Rated Active Power	MW
Imax	Rated DC-Current	kA
tapnom	Nominal Turns-Ratio (t2/t1)	
alphanom	Nominal Firing Angle	deg
i_diode	Diode-/Thyristor Converter	
i_trf	Converter Transformer: Built-In Transformer	
tapmin	Converter Transformer: Minimum Turns-Ratio	p.u.
tapmax	Converter Transformer: Maximum Turns-Ratio	p.u.
Pnold	Losses: No-load losses	kW
swtLossFactor	Losses: Switching loss factor	kW/A
resLossFactor	Losses: Resistive loss factor	Ohm
Rthy	Thyristor-Resistance (at On)	Ohm
Goff	Thyristor-Conductance (at Off)	S
Gs	Snubber-Conductance	S
Cs	Snubber-Capacity	uF
rres	Rectifier resistance (DC-side)	mOhm
rind	Rectifier inductance (DC-side)	uH
fr_way	ANSI/IEEE Parameters	
CCCconv	Capacitor commutated converter (CCC)	
Ccap	Capacitance (CCC)	uF
frame_mov	Metal Oxide Varistor (only for CCC)	

B Signal Definitions

Table B.1: Input/Output signals

Name	Description	Unit	Type	Model
alpha	Firing Angle	rad	IN	RMS, EMT
tap	Tap-Position		IN	RMS, EMT
short_dc	DC Bypass		IN	RMS, EMT
block_all	AC Blocking		IN	RMS, EMT
fref	Reference Frequency	p.u.	IN	RMS
Fmeas	Frequency	Hz	IN	EMT
gamma	Extinction Angle	rad	OUT	RMS, EMT
gamma_min	Extinction Angle (Min. in one cycle)	rad	OUT	RMS, EMT
gamma_state	Gamma State		OUT	RMS
Ip_A	Thyristor Current (pos Thyristor, Phase A)	kA	OUT	EMT
Ip_B	Thyristor Current (pos Thyristor, Phase B)	kA	OUT	EMT
Ip_C	Thyristor Current (pos Thyristor, Phase C)	kA	OUT	EMT
Im_A	Thyristor Current (neg Thyristor, Phase A)	kA	OUT	EMT
Im_B	Thyristor Current (neg Thyristor, Phase B)	kA	OUT	EMT
Im_C	Thyristor Current (neg Thyristor, Phase C)	kA	OUT	EMT
Upc_A	Capacitive Voltage (pos Snubber Capacity, Phase A)	kV	OUT	EMT
Upc_B	Capacitive Voltage (pos Snubber Capacity, Phase B)	kV	OUT	EMT
Upc_C	Capacitive Voltage (pos Snubber Capacity, Phase C)	kV	OUT	EMT
Umc_A	Capacitive Voltage (neg Snubber Capacity, Phase A)	KV	OUT	EMT
Umc_B	Capacitive Voltage (neg Snubber Capacity, Phase B)	kV	OUT	EMT
Umc_C	Capacitive Voltage (neg Snubber Capacity, Phase C)	kV	OUT	EMT
U0	Zero Sequence Voltage	kV	OUT	EMT
Uc_A	Capacitor voltage (CCC), phase A	kV	XI	EMT
Uc_B	Capacitor voltage (CCC), phase B	kV	XI	EMT
Uc_C	Capacitor voltage (CCC), phase C	kV	XI	EMT

C References

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[3] J. Reeve, J. A. Baron, and G. A. Hanley. A technical assessment of artificial commutation of hvdc converters with series capacitors. *IEEE Transactions on Power Apparatus and Systems*, 87(10):1830–1840, 1968.

[4] Working Group B4.34. Capacitor Commutated Converters (CCC) HVDC Interconnections - Digital Modeling and Benchmark Circuit. Technical report, CIGRE, June 2008.

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